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```
mysql> CREATE TABLE roles(role_id INT PRIMARY KEY, role_name VARCHAR(50));
Query OK, 0 rows affected (0.10 sec)
```

```
mysql> DESCRIBE roles;
```

Field	Type	Null	Key	Default	Extra
role_id	int	NO	PRI	NULL	
role_name	varchar(50)	YES		NULL	

2 rows in set (0.04 sec)

```
mysql> CREATE TABLE users(user_id INT PRIMARY KEY,name VARCHAR(50),email VARCHAR(50),role_id INT);
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DESCRIBE users;
```

Field	Type	Null	Key	Default	Extra
user_id	int	NO	PRI	NULL	
name	varchar(50)	YES		NULL	
email	varchar(50)	YES		NULL	
role_id	int	YES		NULL	

4 rows in set (0.02 sec)

```
mysql> CREATE TABLE departments(dept_id INT PRIMARY KEY,dept_name VARCHAR(50));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> DESCRIBE departments
-> ^C
```

```
mysql> DESCRIBE departments;
```

Field	Type	Null	Key	Default	Extra
dept_id	int	NO	PRI	NULL	
dept_name	varchar(50)	YES		NULL	

2 rows in set (0.01 sec)

```
mysql> CREATE TABLE employees(emp_id INT PRIMARY KEY,emp_name VARCHAR(50),dept_id INT,manager_id INT,salary DECIMAL(10,2));
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> DESCRIBE employees;
```

Field	Type	Null	Key	Default	Extra
emp_id	int	NO	PRI	NULL	
emp_name	varchar(50)	YES		NULL	
dept_id	int	YES		NULL	
manager_id	int	YES		NULL	
salary	decimal(10,2)	YES		NULL	

5 rows in set (0.01 sec)

```
mysql> CREATE TABLE customers(customer_id INT PRIMARY KEY,customer_name VARCHAR(50),city VARCHAR(50));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> DESCRIBE cusromers;
ERROR 1146 (42S02): Table 'qwe.cusromers' doesn't exist
mysql> DESCRIBE customers;
```

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	
customer_name	varchar(50)	YES		NULL	
city	varchar(50)	YES		NULL	

3 rows in set (0.02 sec)

```
mysql> CREATE TABLE orders(order_id INT PRIMARY KEY,customer_id INT,order_date DATE,total_amount DECIMAL(10,2));
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> DESCRIBE orders;
```

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	
customer_id	int	YES		NULL	
order_date	date	YES		NULL	
total_amount	decimal(10,2)	YES		NULL	

4 rows in set (0.01 sec)

```
mysql> CREATE TABLE accounts(account_no INT PRIMARY KEY,holder_name VARCHAR(50),balance DECIMAL(10,2));
Query OK, 0 rows affected (0.06 sec)
```

```
mysql> DESCRIBE accounts;
```

Field	Type	Null	Key	Default	Extra
account_no	int	NO	PRI	NULL	
holder_name	varchar(50)	YES		NULL	
balance	decimal(10,2)	YES		NULL	

3 rows in set (0.02 sec)

```
mysql> SELECT emp_name,dept_name FROM employees e INNER JOIN departments d ON e.dept_id=d.dept_id;
Empty set (0.07 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id;
Empty set (0.02 sec)
```

```
mysql> SELECT name,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
Empty set (0.02 sec)
```

```
mysql> SELECT name,email,role_name FROM users u LEFT JOIN roles r ON u.role_id=r.role_id UNION SELECT name,email,role_name FROM users u RIGHT JOIN roles r ON u.role_id=r.role_id;
Empty set (0.01 sec)
```

```
mysql> SELECT e.emp_name Employee,m.emp_name Manager FROM employees e LEFT JOIN employees m ON e.manager_id=m.emp_id;
Empty set (0.00 sec)
```

```
mysql> SELECT o.order_id,customer_name,city,total_amount FROM customers c JOIN orders o ON c.customer_id=o.customer_id;
Empty set (0.01 sec)
```

```
mysql> SELECT dept_name,COUNT(*),AVG(salary),MAX(salary) FROM employees e JOIN departments d ON e.dept_id=d.dept_id GROUP BY dept_name;
Empty set (0.03 sec)
```

```
mysql> SELECT emp_name,salary,dept_name FROM employees e JOIN departments d ON e.dept_id=d.dept_id WHERE salary>50000 AND dept_name='CSE';
Empty set (0.02 sec)
```

```
mysql> SELECT customer_name,SUM(total_amount) TotalPurchase FROM customers c JOIN orders o ON c.customer_id=o.customer_id GROUP BY customer_name ORDER BY TotalPurchase DESC;
Empty set (0.00 sec)
```

```
mysql> CREATE TABLE students (
->     student_id INT PRIMARY KEY,
->     student_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.09 sec)

```
mysql> DESCRIBE students;
```

Field	Type	Null	Key	Default	Extra
student_id	int	NO	PRI	NULL	
student_name	varchar(50)	YES		NULL	

2 rows in set (0.03 sec)

```
mysql> CREATE TABLE faculty (
->     faculty_id INT PRIMARY KEY,
->     faculty_name VARCHAR(50)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> DESCRIBE faculty;
```

Field	Type	Null	Key	Default	Extra
faculty_id	int	NO	PRI	NULL	
faculty_name	varchar(50)	YES		NULL	

2 rows in set (0.00 sec)

```
mysql> CREATE TABLE courses (
->     course_id INT PRIMARY KEY,
->     course_name VARCHAR(50),
->     faculty_id INT,
->     FOREIGN KEY (faculty_id) REFERENCES faculty(faculty_id)
-> );
```

Query OK, 0 rows affected (0.08 sec)

```
mysql> DESCRIBE courses;
```

Field	Type	Null	Key	Default	Extra
course_id	int	NO	PRI	NULL	
course_name	varchar(50)	YES		NULL	
faculty_id	int	YES	MUL	NULL	

3 rows in set (0.02 sec)

```
mysql> CREATE TABLE enrollments (
->     student_id INT,
->     course_id INT,
->     FOREIGN KEY (student_id) REFERENCES students(student_id),
->     FOREIGN KEY (course_id) REFERENCES courses(course_id)
-> );
```

Query OK, 0 rows affected (0.07 sec)

```
mysql> DESCRIBE enrollments;
```

Field	Type	Null	Key	Default	Extra
student_id	int	YES	MUL	NULL	
course_id	int	YES	MUL	NULL	

2 rows in set (0.02 sec)

```
mysql> SELECT student_name,course_name,faculty_name FROM students s JOIN enrollments e ON s.student_id=e.student_id JOIN courses c ON e.course_id=c.course_id JOIN faculty f ON c.faculty_id=f.faculty_id;
Empty set (0.02 sec)
```

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(104,'Anil',25000); COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

Query OK, 1 row affected (0.02 sec)

Query OK, 0 rows affected (0.00 sec)

```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
104	Anil	25000.00

1 row in set (0.00 sec)

```
mysql> START TRANSACTION; INSERT INTO accounts VALUES(105,'Ravi',10000); ROLLBACK;
Query OK, 0 rows affected (0.01 sec)
```

```
Query OK, 1 row affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
104	Anil	25000.00

```
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-5000 WHERE account_no=101; UPDATE accounts SET balance=balance+5000 WHERE account_no=102; COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.01 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM account;
ERROR 1146 (42S02): Table 'num.account' doesn't exist
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
104	Anil	25000.00

```
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance-3000 WHERE account_no=101; UPDATE accounts SET balance=balance+3000 WHERE account_no=999; ROLLBACK;
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
104	Anil	25000.00

```
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION; UPDATE accounts SET balance=balance+2000 WHERE account_no=101; SAVEPOINT s1; UPDATE accounts SET balance=balance-1000 WHERE account_no=101; ROLLBACK TO s1; COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.02 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
Rows matched: 0 Changed: 0 Warnings: 0
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM accounts;
```

account_no	holder_name	balance
104	Anil	25000.00

```
1 row in set (0.00 sec)
```

```
mysql> show tables;
```

Tables_in_num
accounts
courses
customers
departments
employees
enrollments
faculty
orders
roles
students
users

```
11 rows in set (0.06 sec)
```